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Ploughshares to Swords

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Since the end of the Second World War, the American arms industry has morphed into an exclusive, but lucrative market. The growing demands of a modern military require a set of specialized, and heavily concentrated companies. This, with their growing role in politics and academia, forms the basis of the Military-Industrial Complex

In 1961, President Eisenhower, himself the Supreme Commander of Allied Forces in North Africa during the Second World War, cautioned against a new threat in his farewell speech (1). While his warnings remained vague, he was likely referring to an unsettling trend which was set at the beginning of his presidency. In July 1953, the Korean War (technically) ended and the US, in an unprecedented move, maintained a significant standing army (2). It did so with its eyes West of the peninsula, to its former ally, the Soviet Union. The defence establishment prepared itself for a war against the most powerful foe it would ever face. A war which would require a new approach to arms development.

During the Second World War, the pinnacle of industrialized conflict, materiel was developed to meet the demands of forces deployed on an active battlefield. During the frozen conflict of the mid-century in the age of deterrence and limited conflicts across the globe, the objectives of the US military, much less the supplies it needed to support those objectives, became less clear. At the same time, US command shifted away from the WWII strategy of mass production and towards one of technological superiority. The thinking was that since the Soviet Union dominated in sheer volume, the US had to counterbalance with a smaller number of advanced weapons (3).

Those advanced weapons required more specialized companies. A report, published by RAND a year after Eisenhower's speech, noted that in WWII, the biggest players (often referred to as "primes") consisted mostly of automotive, steel, and chemical companies. General Motors, Ford, Bethlehem Steel, and Chrysler all sat among the top ten in percent of active contracts. By 1961, none of them, save General Motors at number nineteen, would even make the list. They were instead replaced by General Dynamics, Lockheed, Martin (who were still separate at the time), Boeing, and General Electric, with up and comers Raytheon and IBM making their first appearances (4). The modern military no longer simply needed bullets, bread, steel, and fire, it required advanced avionics, complex computing machines, and sophisticated aircraft. At the time, RAND remarked that this shakeup demonstrated the volatility of the defence market, yet the fact that these companies remain the top defence contractors today is a testament to the staying power of these firms dealing in specialized knowledge.

Therefore, the need to provide for an advanced, professional standing army dramatically transformed the defence industry from the supplier of materiel to an alliance between the military, the defence industry, and politicians to dramatically increase defence spending. This phenomenon would be, as termed by Eisenhower in his speech, “the military-industrial-complex.”

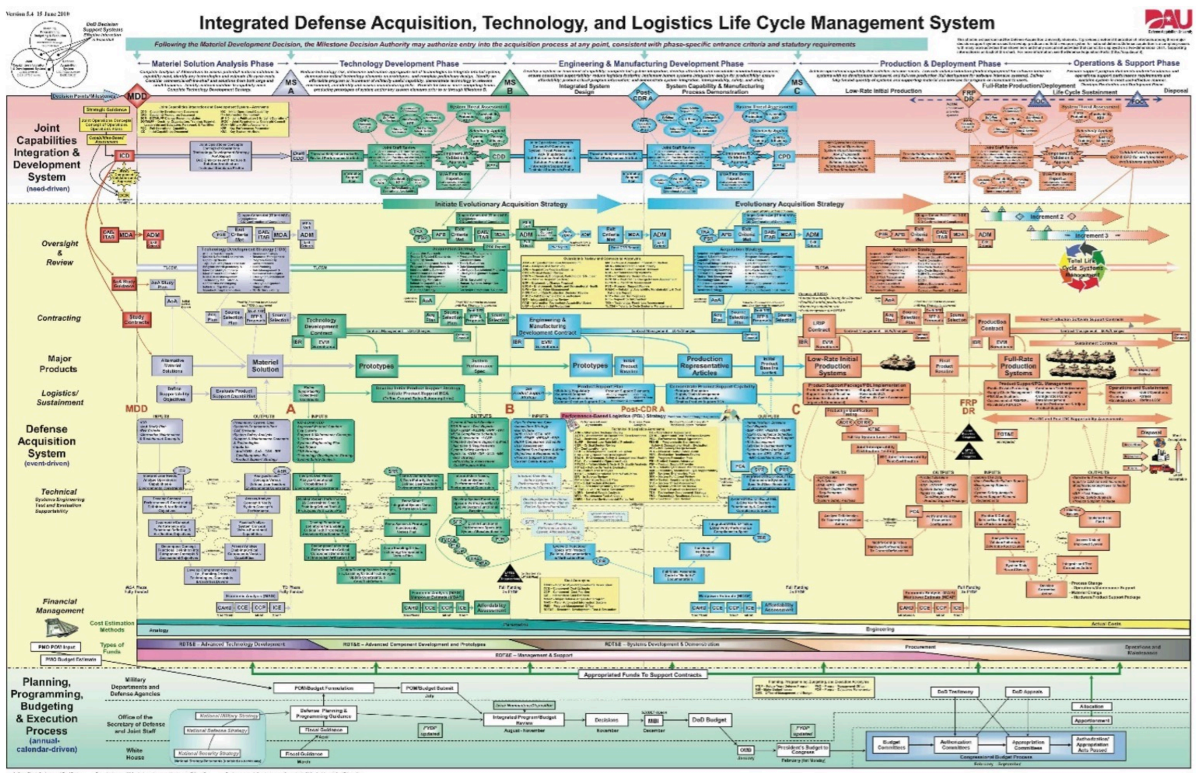
However, at its core, the development and production of supplies comes down to a much less impressive, but critical, process: defence acquisition and procurement.

This paper does not serve as an exhaustive analysis of the defence industry. Indeed, there is an entire discipline of defence economics dedicated to how states fund their militaries and the various inefficiencies therein. However, this should provide a brief overview of some of the unique characteristics of the modern defence market to illuminate the broad issues with defence spending across the globe. As this acts as a companion to a later paper on Canada’s military

modernization, it also serves to highlight why copying the United States’ model is not the clear answer to strengthening Canada’s national defence.

Buyers and Sellers

Defence acquisition and procurement is the means through which militaries guide research and development (R&D) and purchase supplies from producers. That's it. In theory it is quite simple, a state allocates money to a defence budget and that money (oftentimes a great deal of it) is spent on ammunition, tanks, trucks and everything else a military might need to conduct operations. In reality, the process looks something like this:



Arguably one of the less complicated government org charts (5).

We will obviously avoid getting into the bureaucracy of the MIC, but the point here is that it is indeed incredibly bureaucratic. Strategic specialists in military intelligence identify needs for which engineers develop solutions and pitch them to procurement officers. Potential solutions are tested rigorously along a variety of criterion before contracts are eventually awarded to the "winner." Once the contract is awarded, the developer is privy to specialized knowledge and access to materials which restrict the government's ability to look for new contractors should issue arise down the line (6).

Another key characteristic of the MIC is that its very small. The push towards highly advanced systems restricted the number of companies that could participate in defence contracting. Likewise, the government as the sole buyer means they have complete control over which suppliers enter and exit the market. At the same time, as a public body the government is subject to numerous regulatory controls which, through a quirk of the contract renegotiation process, can limit the profitability of defence contracts (7). As a result, it is technical estimates, not a competitive market, which determines the price of goods (8). Thus, the buyer can never be sure that they are getting its good at its lowest cost, and the seller cannot shop around for a better deal. The result means the system is rife with inefficiency. The modern MIC would only become smaller after the Cold War. In 1993, President Clinton's defence secretary ordered the MIC to be consolidated to trim the fat that had been collected since the 1950s. The number of defence contractors who dealt directly with the Department of Defence subsequently reduced from over 50 to six (9).

Pair this with the prevalence of uncertainty. In 1980, J. S. Glasner released a book criticizing the inefficiencies in the defence industry caused, in part, by overly expensive weapons (10). A more recent paper using models found that when future threats to national defence are unclear, procurement decisions tend to be biased towards developing more "high-quality" weapons leading to inefficiencies and unforeseen complications in the field (11). In the national security, uncertainty abounds. Equipment (especially planes and ships) take a long time to produce, and risk becoming obsolete by the time they are deployed, so developers have an incentive to stay as far ahead as possible. On top of this, the military's environment and adversaries are never certain, causing nations to occasionally build up arsenals which have little use and ultimately waste resources. To rectify this, developers have increasingly tried to create platforms which could be adapted to perform a variety of missions across theatres. However, as will be noted in subsequent examples, in doing so they can become more difficult to operate and lead to higher failure rates, all while failing to satisfy the requirements of any of its roles. It is this “jack of all trades” mindset that produced controversial developments such as the Littoral Combat Ship, Bradley Infantry Fighting Vehicle and V-22 Osprey.

Then there are issues with the products themselves. Their “allure” which Ron Smith claims is not unlike government megaprojects. Engineers jump at the chance to design something unique, politicians want to flex their nation’s power, and workers look forward to the jobs the project will create (12). All of these parties can take advantage of the government as a political buyer and lobby for decisions that deviate from the economic rationale (13). In short, people like building cool things and the temptation to do so might obscure some of the issues that lead to egregious cost overruns.

Small, tightly controlled markets, painfully bureaucratic processes, the inability to know all the relevant information, and emotional attachment mean the contemporary procurement system is bound to misfire. Indeed, there are numerous examples of notorious investments, but there are other considerations beyond simply the market structure.

First, while weapon developments are initiated to meet a specific need for the armed forces, those needs are not always reflective of the exact circumstances the military might face. This reliance on a top-down development approach persists to today, Take the infamous “Littoral Combat Ship” which the US Navy has been forced to buy. Its modular design and overreliance on technology are not well suited for the long-range, fleet-based operations that the US often embarks on.

Second, by simply looking at the diagrams explaining the process, one might assume that arms procurement is a mechanical, scientific approach. Yet it is often far messier, with competing interests and politics muddying what should be standard procedures. Thus, despite its focus on the economics of procurement, this paper is not titled as such. Procurement pertains to the interaction between contractors and defence officials ina closed market. Yet all these interactions exist within the larger Military-Industrial Complex, where efficiencies in production and performance can often be set aside for political gain.

Finally, while it can be easy to point and laugh at the tone-deaf equipment that makes it into soldiers' hands and the cast of characters who got them there. But when these weapons fail to perform, they often result in the deaths of those who use them.

Failing to perform rigorous tests on the M-16 rifle, the Ordnance Corps allowed it to get it into the hands of soldiers in Vietnam where it became notorious for jamming and led to the deaths of countless marines (14).

The issue persists even in peacetime. The V-22 Osprey tiltrotor was intended to be a hybrid between a plane and a helicopter. Instead, it fulfils neither role and has resulted in the death of (at the time of writing) sixty-two marines, none of which is the result of enemy action. It can be easy to forget that for defence contractors, failure incurs an additional cost. Any time a new platform is deployed it has the potential to kill people. As such, even though a modern military might look good on paper, pushing the deployment of cutting-edge equipment without a particularly good reason risks causing unnecessary deaths.

The Military-Industrial Complex today

After the Clinton merger, Lockheed and Martin merged, consolidating an already tiny market that increased the suppliers' bargaining power. Although many critics of the modern MIC complain that this means contractors can charge exorbitant fees, those aforementioned contractual regulations do mitigate the worst possibilities of a tiny market. That being said, due to the cost plus fixed-fee (CPFF) contracts they have with the government, taxpayers cover their development costs. This can incentivize more ambitious projects by developers but leave the government on the hook for cost overruns, even if development goes on for decades without any product to show for it (15). As a result, the average US citizen pays approximately a thousand dollars (USD) on defence per year (16). With lacklustre social services, US citizens question how so many inefficiencies can be allowed to continue.

In a world of growing instability, Canada's outdated military sticks out like a sore thumb among NATO's wealthy members. In a forthcoming paper, we hope to address the challenges faced by Canada's procurement system. As such, this paper acts as a brief primer on some (but hardly all) of the obstacles that all nations must contend with to become a world-class military power.

However, this essay only looks at the MIC as it interacts with the military during the procurement process and some of the characteristics of its unique market structure. The Military-Industrial Complex has far-reaching implications for the commercial economy and society at large. As James Carrol writes in *House of War* "Much of the postwar boom that institutionalized the wealth of the United States was driven by the Engines of the Pentagon" (17). Among one of the consequences of the MIC is that household names now produce killing machines. Samsung makes washing machines and televisions, as well as self-propelled artillery and autonomous machine guns. Toshiba develops, among other things, surface-to-air missiles. Dow Chemical, which makes Styrofoam and Saran Wrap, also supplied the US military with Agent Orange, the chemical defoliant deployed over Vietnam. As was the sentiment during the Vietnam War, by buying consumer goods ordinary citizens feel complicit in the violence these companies help to perpetuate. At the same time, the relationship between academia and the defence industry has deepened. According to a Quincy Institute brief published last year, at least 77 percent of American foreign-policy think-tanks have received some funding from defence contractors (18). The formidable Military-Industrial Complex is not simply a market for exchanging goods, but has truly developed into a phenomenon which reaches into all facets of society.

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